DSC 401

Roll No. 24BCD035

B.Tech (Computer Science & Engg.) (Data Science) 4th Semester Examinations 2025-26

Data Mining

Paper Code: DSC 401

Maximum Marks: 45

Maximum Time: 3 hr

(Write your Roll No. on the top immediately on receipt of this question paper)

Note: Attempt any two parts from each question. **Assume suitable data, if necessary.**

S.No.	Questions	Marks	BL/CO
1(a)	Explain the steps involved in the data mining process. Explain why data preprocessing is an important step?	4.5	2/2
1(b)	Apply Z-Score normalization techniques on the following dataset. [3500, 4200, 4700, 5200, 5500, 4600, 3700, 10000] Show the normalized dataset.	4.5	3/2
1(c)	A company collects data on 1000 customers, recording their weekly spending data on groceries, electronics, clothing, travel, and entertainment. A Data Scientist plans to apply Principal Component Analysis (PCA) to this dataset. (a) Show graphically, how the spending data for each customer will be represented in the new dataset. (b) How does this new representation help in customer segmentation.	4.5	4/2
2(a)	Given the following transaction dataset,	4.5	3/3

Transaction ID	Items
T1	A, B, D
T2	A, B, C, D
T3	C, D
T4	B, C, D
T5	A, B, C
T6	B, C
T7	A, B, C, E
T8	A, B, E

	(a) Generate all frequent itemsets using Apriori algorithm with minimum support = 40%. Show stepwise candidate generation and pruning. (b) From the frequent itemsets, generate all association rules with minimum confidence = 80%.		
2(b)	For the following dataset, with minimum support = 60%. **TID** | **Items** *T1* | A, B, C *T2* | A, C, D *T3* | B, D *T4* | A, B, C *T5* | B, C, E Use FP-Growth algorithm to construct FP-tree. Show each step clearly.	4.5	4/3
2(c)	Given the following conditional pattern base for item 'C': {(A, B: 3), (A: 2), (B: 2)} *(i)* Construct the conditional FP-tree for item 'C' (if required). *(ii)* Generate all frequent itemsets for 'C' with a minimum support count of 2, clearly showing how the frequent itemsets are obtained from the pattern base.	4.5	4/3
3(a)	What is regression? For the following training data, **X** | 4 | 5 | 7 | 8 | 10 **Y-true** | 8 | 6 | 7 | 10 | 12 Derive the Mean Squared Loss Function for the linear regression function and calculate training loss.	4.5	3/4
3(b)	Given the following training data, **X1** | 1 | 2 | 5 | 6 | 3 **X2** | 1 | 2 | 6 | 6 | 4 **Y** | 1.2 | 2.5 | 7.3 | 9.5 | ? Apply the k-NN regression algorithm and predict the target for (X1=3, X2=4). Show the necessary calculations.	4.5	3/4

3(c)	Given the following true and predicted data, **True Class** 1 1 1 1 1 0 0 0 0 0 **Predicted class** 1 1 0 1 0 1 1 0 1 0 Draw the confusion matrix and calculate the false positives, false negatives, precision and recall.	4.5	3/4
4(a)	Explain the process of constructing a decision tree from a training dataset. Describe how attribute selection measure Information Gain is used at each step to determine the optimal splitting criterion.	4.5	2/4
4(b)	A bank wants to predict whether a credit card holder will default on a loan based on their credit score X. A logistic regression model has been trained with the following parameters $\theta_0 = -6, \theta_1 = 2.4$ (*i*) Calculate the probability that a credit card holder will default if their credit score X = 3. (*ii*) Calculate the probability that a credit card holder will default if their credit score X = 6. (*iii*) Is linear regression model also suitable for predicting the probability of defaulter, justify your answer.	4.5	5/4
4(c)	A hospital wants to predict whether a patient has Pneumonia or not based on three binary features. The training dataset of the patient is as follows:	4.5	4/4

Candidate	Cold	Fever	Cough	Pneumonia
1	No	No	No	No
2	Yes	Yes	No	Yes
3	No	Yes	Yes	No
4	No	No	Yes	No
5	No	Yes	No	No
6	Yes	Yes	Yes	Yes
7	No	No	No	No
8	Yes	No	No	Yes

	Using Naive Bayes, determine whether the patient who has cold, do not have fever but have cough will have Pneumonia or not.		
5(a)	Given the data points: A = (3,11) | B = (1,4) | C = (9,3) | D = (6,8) | E = (5,5), | F = (6,4) and initial centroids, $C_1 = (1,5)$, $C_2 = (5,10)$, with K = 2, (*i*) Perform one iteration of K-means and update the centroids. (*ii*) Does the initial choice of centroids affect the final clusters, explain?	4.5	5/5
5(b)	Given the 2D points A (1,1) | B (2,1) | C (4,2) | D (8,8) | E (8,9) | F (25,80) Use DBSCAN algorithm with parameters $\varepsilon = 2$ and MinPts = 2 (*i*) Identify which points are core points, border points, and noise points. (*ii*) Find the clusters.	4.5	3/5
5(c)	Explain AGNES with an example (≥6 points). Construct the dendrogram step-by-step and describe how clusters are determined. How does the cut-off height affect the clustering?	4.5	4/5